Seungmin Oh 오승민

Ph.D. candidate in Artificial Intelligence

Multi-Modal Artificial Intelligence Lab, Ajou University, South Korea

Advised by Prof. Jongbin Ryu, Ph.D. expected Feb 2028

seungmin.oh16@gmail.com
soonge.github.io
Google Scholar
github.com/SoongE
linkedin.com/in/soonge
Suwon, South Korea

Ph.D. candidate building **efficient AI** for deployment under tight compute limits, from edge devices to robot control loops, across vision, LLM, VLM, and VLA models.

Education

Integrated M.S./Ph.D. in Artificial Intelligence — Feb 2022 - present
Ajou University, Suwon, South Korea, advised by Prof. Jongbin Ryu

B.S. in Software and Computer Engineering — Feb 2016 - Feb 2022
Ajou University, Suwon, South Korea

Research experience

Multi-Modal Artificial Intelligence Lab, Ajou University — Feb 2022 - present
Graduate Researcher, advised by Prof. Jongbin Ryu

Vision-language-action models for robotic manipulation — 2025 - present

- Training methodology: world-state change prediction instead of raw actions, preserving VLM grounding (in prep.).
- Inference acceleration: fast action controller with lazy gating; full VLA invoked only on anticipated failure (in prep.).
- Failure detection: early visual observer for semi-closed-loop action-chunk execution [S.2].

Efficient LLM compression — 2024 - present

- Structured pruning: overcomplete training for scaled recovery capacity, up to +8.4 points at zero deployment cost [C.3].
- Structured pruning: curriculum-based layer-wise compression schedule, 1.7× faster convergence at half the GPU memory [C.4].

Efficient vision and test-time adaptation — 2022 - present

- On-device classification: 1st place, 2025 IEEE Low-Power Computer Vision Challenge Track 1 [A.1].
- Test-time adaptation: candidate verification and divergence shift [S.1].
- Re-parameterization: branch-level neural substitution [C.1].
- Representation learning: transformation-aware prompt conditioning via re-calibrated contrastive loss [C.2].
- Medical: CT-based triage for small-bowel obstruction [J.1]; CNN vs. ViT architectural behavior [J.2].

Funded applied research projects (ETRI, NIA, 42dot) — 2021 - 2024

- Real-time detection transformer for autonomous driving (42dot); source-free domain adaptation for on-device face recognition (ETRI); lymphoma recognition and survival analysis from 3D CT (NIA); masked-face recognition with domain adaptation (ETRI).

GGQ Company Inc., Seoul, South Korea — 2020
Research Intern

- Game-scene understanding: player-name, skill, and HUD OCR in League of Legends; face recognition for attendance.

Publications

S under review, C conference, A preprint, J journal, P patent

- [S.2] Donghyun Kim, **Seungmin Oh**, Dooyeon Na, Jongbin Ryu. Who Watches the VLA? Early Visual Observer for Semi-Closed-Loop Action-Chunked Execution. *Under review*.
- [S.1] **Seungmin Oh**, Seunghun Kang, Jongbin Ryu. Efficient Test-Time Adaptation through Candidate Verification and Divergence Shifts. *Under review*.
- [C.4] Donggeon Lee, Dooyeon Na, **Seungmin Oh**, Jongbin Ryu. Layer-wise Curriculum Learning for Efficient LLM Compression. *EMNLP*, 2026.
- [C.3] **Seungmin Oh**, Donggeon Lee, Jongbin Ryu. Train Overcomplete, Deploy Compact: Scaling Recovery Capacity for Structured LLM Pruning. *EMNLP*, 2026.
- [C.2] **Seungmin Oh**, Seunghun Kang, Jongbin Ryu. Re-calibrated Contrastive Loss for Transformation-Aware Prompt Conditioning in Vision-Language Models. *BMVC*, 2026.
- [A.1] Zihao Ye et al., including **Seungmin Oh**, Jongbin Ryu. Evaluation of Winning Solutions of 2025 Low Power Computer Vision Challenge. *arXiv preprint*, 2025.
- [C.1] **Seungmin Oh**, Jongbin Ryu. Neural Substitution for Branch-Level Network Re-parameterization. *ACCV*, 2024.
- [J.2] **Seungmin Oh**, Namkug Kim, Jongbin Ryu. Analyzing to Discover Origins of CNNs and ViT Architectures in Medical Images. *Scientific Reports*, 14:8755, 2024.

[J.1] **Seungmin Oh**, Jongbin Ryu, Hojung Shin, Jeongho Song, Sangyong Son, Hoon Hur, Sanguk Han. Deep Learning Using Computed Tomography to Identify High-Risk Patients for Acute Small Bowel Obstruction: Development and Validation of a Prediction Model. *International Journal of Surgery*, 109(12), 2023 (IF 15.3, JCR top 1%).

[P.2] Jongbin Ryu, Dooyeon Na, **Seungmin Oh**, Donggeon Lee. System and Method for Compressing a Deep Learning Model Based on Layer-Wise Curriculum Learning. *Korean Patent Application No. 10-2026-0123632*, filed Jul 2026.

[P.1] Hojung Shin, Jongbin Ryu, **Seungmin Oh**. Electronic Device and Method for Classifying Small Bowel Obstruction Using Abdominal Computed Tomography Images. *PCT International Patent Application No. PCT/KR2024/014604*, filed Sep 2024.

Honors and awards

IEEE Low-Power Computer Vision Challenge, Winner Apr 2025
CVPR 2025 Workshops, organized by IEEE, CVF, Qualcomm, and Purdue University; ranked 1st among 45 teams
Layer Imitation: Efficient Vision-Language Model on the Mobile NPU

Google Solution Challenge, Global Top 100 May 2023
Google
'Smiler,' an expression-training app for children with ASD

ICT Mentoring Conference, Excellence Award Dec 2021
ACK, KOSIA
Recommender System for Web Search Based on NLP to Improve User Search Environment

Student Paper Competition, First Prize Jul 2021
Korea Digital Contents Society
Lightweight Video Search Using Video-to-Text Generation and Retrieval

Research funding

Doctoral Research Grant Sep 2025 - Aug 2027
National Research Foundation of Korea (NRF), $36K
Progressive Linear Activation for Depth-wise Reparameterization of Large-scale Models

Lesion-Class Re-labeling in Medical Image Training Data Oct 2022 - Dec 2023
National Information Society Agency (NIA), $20K

Technical skills

Infrastructure	Built and operate a multi-node GPU cluster (about 20 nodes, 8× RTX 3090 each): containerized environment provisioning, per-node CUDA isolation, unattended long-running training pipelines.
Programming	Python; C/C++ with SIMD- and cache-aware performance optimization; CUDA.
ML tooling	PyTorch, Hugging Face, Weights & Biases, DVC, Docker, Kubernetes, Git and GitHub Actions.
On-device	Qualcomm AI Hub (Snapdragon NPU); structured pruning and re-parameterization for compact deployment.

Teaching and leadership

Lecturer, tutorials for undergraduate AI research assistants, Ajou University (3 terms) 2023 - 2024
Teaching Assistant, Capstone Design Project, Ajou University 2023
Founding Member, Google Developer Student Club, Ajou University chapter Sep 2022 - Dec 2023